determine the membership. Dynamic groups can be useful when groups become very large because they avoid having to explicitly list every member of the group but can cause performance problems as the server needs to perform potentially expensive search operations to determine the members of a group.

To find which static groups a user is a member of in LDAP, you must perform a search against the directory for all groups that have that user's *distinguished name* as a value of their `member` attribute, as shown in listing 8.3. First you need to connect to the LDAP server using the Java Naming and Directory Interface (JNDI) or another LDAP client library. Normal LDAP users typically are not permitted to run searches, so you should use a separate JNDI `InitialDirContext` for looking up a user's groups, configured to use a connection user that has appropriate permissions. To find the groups that a user is in, you can use the following search filter, which finds all LDAP `groupOfNames` entries that contain the given user as a member:

```
(&(objectClass=groupOfNames)(member=uid=test,dc=example,dc=org))
```

To avoid LDAP injection vulnerabilities (chapter 2), you can use the facilities in JNDI to let search filters have parameters. JNDI will then make sure that any user input in these parameters is properly escaped before passing it to the LDAP directory. To make use of this, replace the user input in the field with a numbered parameter (starting at 0) in the form `{0}` or `{1}` or `{2}`, and so on, and then supply an `Object` array with the actual arguments to the `search` method.